

Automated Temperature-Based Fan Control System Using Arduino UNO

Project 1 • Working Prototype • Version 1.0

Abstract

An automatic fan-control system was developed using an Arduino UNO and DHT11 temperature sensor. The Arduino reads temperature and generates PWM on D9 to control a 12 V, 0.15 A DC fan through an IRLZ44N MOSFET. The fan uses a separate 12 V, 1 A adapter, with a common ground between the power system and Arduino.

Problem & Objective

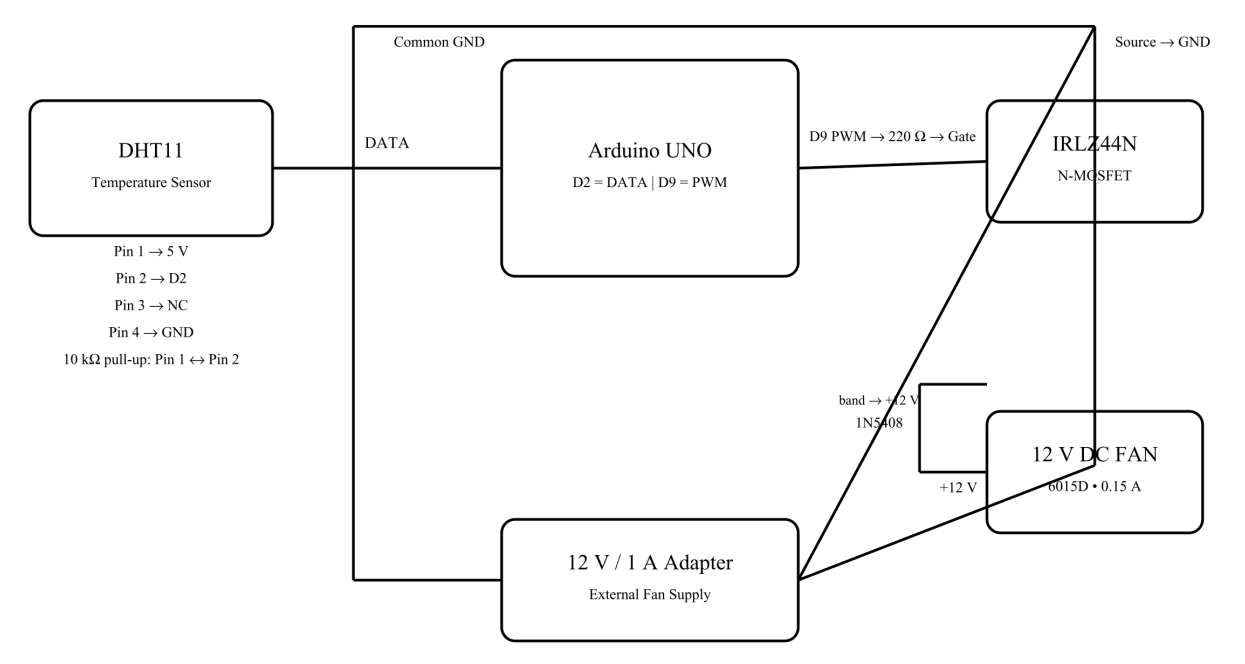
Fixed-speed/manual fan operation does not adapt to changing temperature. The objective is to demonstrate a sensor-to-actuator embedded control system that measures temperature, makes a programmed decision, and automatically controls a real electrical load.

Hardware

Component	Specification / role
Arduino UNO	Main controller; D2 sensor input, D9 PWM output
DHT11	Temperature/humidity sensor
6015D fan	12 V, 0.15 A cooling load
IRLZ44N	N-channel MOSFET for low-side switching/PWM
1N5408	Flyback protection diode
Resistors	220 Ω gate resistor; 10 k Ω gate pull-down; 10 k Ω DHT11 pull-up
Adapter	12 V DC, 1 A fan supply

Circuit & Connections

Automated Temperature-Based Fan Control System — Circuit Diagram



DHT11: Pin 1→5 V, Pin 2→D2, Pin 3→NC, Pin 4→GND; 10 k Ω pull-up between VCC and DATA. **Fan control:** D9→220 Ω →MOSFET Gate; 10 k Ω Gate→GND; Source→common GND; Fan negative→Drain; fan positive→+12 V. Adapter negative and Arduino GND are common. 1N5408 is across the fan with its banded end toward +12 V.

Working & Control Logic

The DHT11 measures temperature. The Arduino compares the reading with preset thresholds and changes PWM duty cycle on D9: **<30 °C = OFF**; **30–30.9 °C = LOW (80/255)**; **31–31.9 °C = MEDIUM (160/255)**; **≥32 °C = HIGH (255/255)**. The MOSFET switches the externally powered 12 V fan.

Testing & Result

Temperature	Fan state	PWM
<30 °C	OFF	0/255
30–30.9 °C	LOW	80/255
31–31.9 °C	MEDIUM	160/255
≥32 °C	HIGH	255/255

Initial sensor testing produced approximately 29 °C and 56.4% RH, confirming successful DHT11 communication. The temperature-dependent fan-control program was subsequently operated successfully. Add measured state-by-state observations to the separate test log.

Problems, Limitations & Future Improvements

The bare DHT11 pin arrangement required verification during assembly. A 1N4007 was unavailable, so a 1N5408 was used. Version 1 has no display, manual button, or data logging. Future versions can add an I²C LCD, status LEDs/buzzer, adjustable thresholds, hysteresis, data logging, and a protective enclosure.

Conclusion

The prototype demonstrates a complete embedded control chain: temperature sensing, microcontroller decision-making, PWM generation, MOSFET switching, and independent DC-load power. It provides a practical foundation for more advanced Arduino and ECE projects.